

RANSHOFEN, Austria

WMO: 110510

Lat: 48.224N

Lon: 13.028E

Elev: 1273

StdP: 14.03

Time Zone: 1.00 (EUC)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.7	17.2	7.4	8.4	13.4	12.8	11.0	18.9	16.2	38.0	14.4	37.5	4.3	100	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	87.5	69.0	83.8	67.6	80.7	66.2	70.2	83.8	68.7	81.0	67.3	78.1	5.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	100.2	74.8	64.5	95.4	73.2	63.3	91.3	72.1	34.9	84.4	33.6	81.3	32.5	78.3	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.1	10.3	DB	7.9	93.4	6.6	2.2	3.1	95.0	-0.8	96.3	-4.5	97.5	-9.3	99.1	(4)
(5)			WB	6.7	73.4	6.5	1.9	2.0	74.8	-1.8	75.9	-5.5	76.9	-10.3	78.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	31.0	34.1	41.6	50.6	58.1	64.4	67.1	66.7	58.4	49.9	39.9	33.8
	DBStd	14.47	7.66	7.85	7.21	7.29	6.75	6.80	6.01	6.02	5.91	6.38	6.61	6.60
	HDD50	2291	589	445	269	79	11	1	0	0	5	82	309	501
	HDD65	5896	1053	864	725	432	230	93	48	53	211	467	754	966
	CDD50	2196	0	1	9	99	263	432	532	517	257	80	6	0
	CDD65	325	0	0	0	1	17	74	115	105	13	0	0	0
	CDH74	2900	0	0	0	33	196	681	995	893	99	3	0	0
	CDH80	828	0	0	0	3	30	180	315	290	10	0	0	0
Wind	WSAvg	4.6	4.9	5.1	5.4	5.0	4.9	4.1	4.0	3.9	4.0	4.4	4.4	4.7
Precipitation	PrecAvg	40.8	2.4	2.1	3.0	2.4	4.2	5.1	5.0	4.7	3.8	2.8	2.7	2.6
	PrecMax	49.8	5.4	6.2	5.7	6.9	7.5	10.1	10.4	10.3	7.6	5.3	6.2	4.6
	PrecMin	29.0	0.2	0.3	0.8	0.0	0.6	1.8	0.7	1.5	1.5	0.5	0.1	0.4
	PrecStd	4.8	1.4	1.4	1.5	1.5	1.9	2.0	2.2	2.0	1.5	1.3	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	57.5	66.5	78.4	83.7	89.5	91.7	92.2	81.0	72.4	61.8	53.2
		MCWB	47.5	48.0	52.4	59.5	66.0	70.3	70.7	69.2	66.5	59.9	53.3	48.0
	2%	DB	48.7	52.0	61.5	72.6	79.4	85.0	87.2	87.2	76.9	66.6	55.7	49.0
		MCWB	44.2	45.2	50.3	57.3	63.9	68.4	69.1	68.6	64.5	57.9	50.2	45.3
	5%	DB	44.7	48.4	57.2	68.7	75.5	81.3	83.3	82.7	72.9	62.9	52.0	45.6
		MCWB	41.7	43.6	48.5	55.4	61.8	66.6	67.5	67.7	62.8	56.5	48.5	42.6
	10%	DB	41.5	44.8	53.3	64.6	71.3	77.9	79.7	78.8	69.2	59.4	49.1	42.7
		MCWB	39.1	40.7	46.3	53.3	59.2	64.7	66.2	65.7	60.8	55.0	46.3	40.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.6	48.9	53.8	61.0	67.9	72.2	72.4	71.7	67.8	61.7	54.0	48.8
		MCDB	52.2	55.7	63.7	74.4	80.9	85.6	86.6	87.1	77.7	68.8	59.9	52.1
	2%	WB	44.7	46.3	51.1	58.2	64.8	69.8	70.1	69.7	65.5	59.3	51.0	45.9
		MCDB	47.8	51.4	59.5	70.7	77.1	82.1	84.3	83.2	74.8	64.8	55.0	48.8
	5%	WB	42.0	43.7	49.1	56.1	62.6	67.9	68.5	68.2	63.7	57.3	48.8	42.9
		MCDB	44.7	47.9	56.3	67.3	72.9	79.0	80.7	80.2	71.2	61.9	51.8	45.0
	10%	WB	39.1	41.2	47.0	54.0	60.4	66.0	66.9	66.7	62.0	55.4	46.4	40.7
		MCDB	41.6	44.6	52.6	63.5	69.7	75.6	77.1	77.1	68.0	59.2	48.9	42.8
Mean Daily Temperature Range	5% DB	MDBR	8.9	11.8	15.5	18.3	17.6	17.8	18.0	18.1	16.2	13.7	9.3	8.4
		MCDBR	13.6	18.7	24.3	26.5	25.6	24.8	25.0	25.6	22.2	20.0	14.9	13.6
		MCWBR	10.1	12.5	14.1	13.0	11.3	9.9	9.4	9.6	11.0	12.0	10.2	10.1
	5% WB	MCDBR	13.0	17.6	22.6	25.1	23.4	22.6	22.6	23.1	20.1	18.1	13.9	12.6
		MCWBR	10.0	12.1	13.6	12.6	10.7	9.4	9.1	9.2	10.2	11.3	9.9	9.8
Clear-Sky Solar Irradiance	taub	0.301	0.326	0.365	0.397	0.406	0.410	0.416	0.410	0.384	0.353	0.326	0.299	
	taud	2.409	2.346	2.303	2.250	2.266	2.292	2.272	2.307	2.349	2.422	2.453	2.444	
	Ebn at Noon	244	262	268	270	272	271	267	263	259	249	233	231	
	Edh at Noon	21	28	35	40	41	40	41	37	33	26	21	18	
All-Sky Solar Radiation	RadAvg	343	639	969	1352	1564	1727	1689	1487	1067	692	363	266	
	RadStd	53	85	117	172	187	159	161	160	141	75	48	35	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (86 neighbors)		+0.83	N/A	N/A	N/A	+0.61	+0.63	N/A	N/A	+110	+45

Nomenclature: See separate page